

UniVerse

A Campus Companion

Room Availability

Real-time room occupancy: which rooms are busy, who's in them, and when each frees up.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Robi (campus explore)

Key files: features/rooms/*

1. What it does

Room Availability shows every room with an OCCUPIED / AVAILABLE chip, the current class (teacher, subject, cohort) with minutes remaining, and the next scheduled class. Like Find Teacher, it's reached from the Explore FAB and derives state from the live `routines` table - no new tables.

2. How occupancy is derived (rooms_controller)

- 1 `fetchAllRoutines()` loads all active routine rows.
- 2 Group rows by room.
- 3 For each room, take today's classes sorted by time.
- 4 If `now` is inside a class window -> **OCCUPIED**: show teacher, subject, cohort, and minutes remaining (end - now).
- 5 Else -> **AVAILABLE**: show the next class today (or 'No more classes').
- 6 Sort rooms by name for a stable list.

This is the exact mirror of Find Teacher - same data, grouped by room instead of by teacher. Both reuse RoutineEntry's startOn/endOn time helpers.

3. Real-time

`streamAllRoutines()` (Supabase Realtime) feeds the controller; when the schedule changes, occupancy recomputes automatically.

4. Likely defense questions

question	answer
How do you know a room is free?	If no class in the timetable covers the current time for that room, it's available; we also show when the next class starts.
Sensors / IoT?	None - it's schedule-based occupancy from the published routine.
Shared logic with Find	Yes - same routines data and time helpers; one groups by teacher, the other by room.
Does it cover all rooms?	Every room that appears in the routine; the room pool itself is managed in Manage Rooms for the engine.